



BY 504: Metabolic Systems Biology

Course Code	BY 504
Course Name	Metabolic Systems Biology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	3rd and 4th year UG and PG
Distribution	Elective
Approval	8th Senate (Oct 2015)

Course Contents

- **Introduction to systems biology and metabolism**
Components of Biological systems (DNA, RNA, Protein, Metabolites), their properties and function. Overview of cellular metabolism, enzyme kinetics and metabolic pathways. Online resources and Tools to study metabolism – KEGG, ECOCYC etc. [6 Lectures]
- **Introduction to -omics with focus on metabolomics**
Biological networks and their significance – at the level of genome, transcriptome, proteome, metabolome and fluxome. Metabolomics - applications and its role in systems biology. Analytical methods for detecting and quantifying metabolites. General work flow and Statistical methods in metabolomics. Pathway and metabolome databases. Case study on metabolomics from literature. [8 Lectures]
- **Metabolic pathways, network reconstruction and constraint based flux analysis**
Pathways of central and secondary metabolism in selected model systems (microbes, plant and animal), Reconstruction of metabolic networks, Stoichiometric matrix. Topological analysis of metabolic network with Elementary flux modes and/or Extreme pathways, Introduction to Constraint based metabolic modelling and Flux Balance analysis. Related software tools and online resources. Case study/studies on Constraint based flux analysis from literature with applications in metabolic engineering and/or drug target identification. [14 Lectures]
- **Introductory ¹³C based fluxomics**
Metabolic Phenotypes, Fundamentals of Metabolic Flux analysis. Current practices of ¹³C Metabolic Flux Analysis – Stable isotope labelling, steady state vs Non-stationary, Isotopomer analysis, Carbon transition networks, mathematical modelling for quantifying fluxes (in-vivo reaction rates), Flux maps. Software tools and online resources. Case study/studies on ¹³C metabolic flux analysis from literature with applications in metabolic engineering and/or understanding metabolic features in diseases such as cancer. [14 Lectures]

References

1. Editor: Bernhard O. Palsson, **Systems Biology: Properties of Reconstructed Networks**, Cambridge University Press, 2006.
2. Choi, Sangdun (Editor) **Introduction to Systems Biology**, Springer Publishers, 2007.
3. J.Nielsen and M.C. Jewett (Editors), **Metabolomics – A powerful Tool in Systems Biology**, Springer Publishers, 2007.
4. Jens O. Krömer, Lars K. Nielsen, Lars M. Blank (Editors), **Metabolic Flux Analysis-Methods and protocols**, Springer Publishers, 2014.
5. Additional reference material and scientific papers will be provided.